Purplle Store Intelligence System

Brigade Road, Bangalore — Store ID: ST1008

End-to-end AI pipeline that processes raw CCTV footage to deliver real-time business metrics: store conversion rate, customer funnel, and anomaly signals.

Project Structure

```
Purple/
├── config/
│   └── config.yaml               # Zone definitions, camera map, operating
hours
├── data/
│   └── Brigade_Bangalore_10_Apr_26.csv   # POS transactions (ground truth)
├── services/
│   ├── api/
│   │   ├── api.py                # FastAPI service: /metrics /funnel /anomalies
/health
│   │   ├── Dockerfile
│   │   └── requirements.txt
│   └── detector/
│       ├── pipeline.py           # CV pipeline: YOLO + CentroidTracker + Redis
events
│       ├── Dockerfile
│       └── requirements.txt
├── tests/
│   └── test_store_intelligence.py  # 30 tests covering unit + integration +
business logic
├── CCTV Footage/
│   ├── CAM 1.mp4  → entry_exit
│   ├── CAM 2.mp4  → makeup
│   ├── CAM 3.mp4  → skincare
│   ├── CAM 4.mp4  → bath_body
│   └── CAM 5.mp4  → hair_fragrance
├── yolov8n.pt                    # YOLOv8 nano model weights
├── docker-compose.yml            # 3-service stack: Redis + detector + api
└── README.md
```

Quick Start

1. Run with Docker (recommended)

```
docker compose up --build -d
```

Services started: | Container | Port | Role | |---|---|---| | `event-broker` | 6379 | Redis Stream event bus | | `store-detector` | — | CCTV pipeline (YOLOv8 + tracker) | | `store-api` | 8000 | REST API |

2. Run Tests (local Python)

```
pytest tests/test_store_intelligence.py -v
# → 30 passed
```

API Endpoints

Endpoint	Description
`GET /health`	Pipeline status + events processed count
`GET /metrics`	Footfall, conversion rate, dwell time, peak hour
`GET /funnel`	4-stage customer funnel with drop-off rates
`GET /anomalies`	Detected crowd surges + unusual entry patterns
`GET /metrics/prom`	Prometheus text format metrics

Example Responses

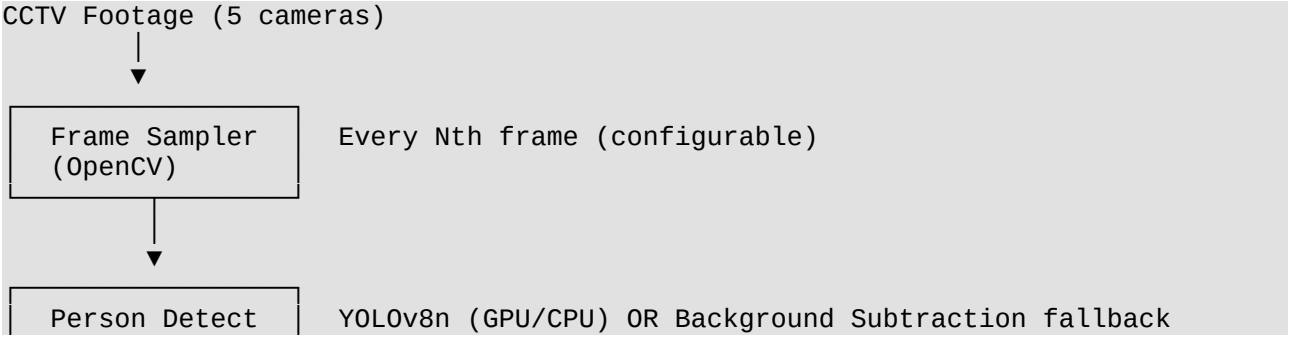
GET /metrics

```
{
  "date": "2026-04-10",
  "store_id": "ST1008",
  "total_footfall": 120,
  "unique_buyers": 24,
  "conversion_rate": 20.0,
  "avg_dwell_time_min": 20.7,
  "peak_hour": 17,
  "staff_count": 5
}
```

GET /funnel

```
{
  "entered": 120,
  "browsed_zone": 96,
  "engaged_salesperson": 58,
  "purchased": 24,
  "drop_off_rates": {
    "browse": 20.0,
    "engage": 39.6,
    "purchase": 58.6
  }
}
```

Pipeline Architecture





Store Context (10-Apr-2026)

Metric	**Value**
Store ID	ST1008
Location	Brigade Road, Bangalore
Operating Hours	12:15 – 21:40
Total Orders	24
GMV	₹44,920
NMV	₹34,831
Active Salespersons	5
Top Categories	Sheet Mask, Lipstick, Body Oil